

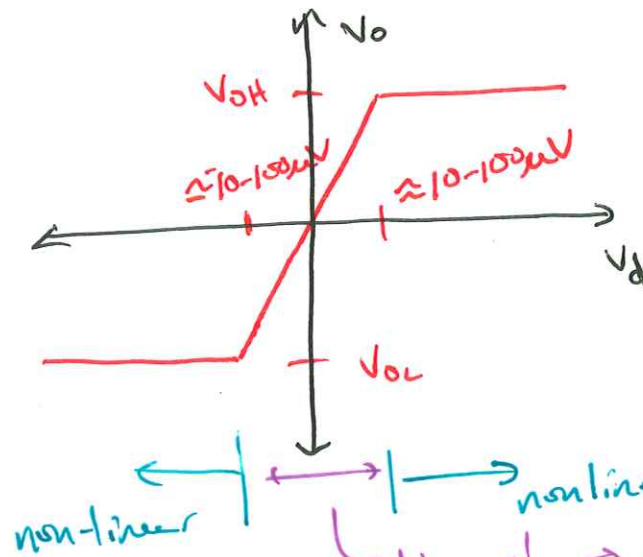
Non-linear Op-Amp Circuits

Present 2 types:

- 1) Use saturation to achieve some goal
- 2) Add non-linear components ...

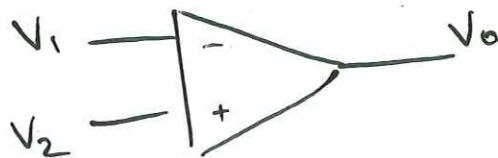
1) Saturation uses

Reminder: →



linear! → only realistically exists when negative feedback exists.

Simplest use case:



comparator!

$$V_o = a(V_2 - V_1)$$

$V_d = V_2 - V_1$, probably not in linear amplification range.

then

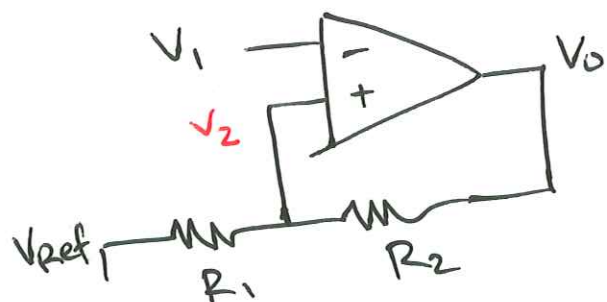
$$V_o = \begin{cases} V_{OL} & V_2 < V_1 \\ ?? & V_2 \approx V_1 \\ V_{OH} & V_2 > V_1 \end{cases}$$



Schmitt Trigger

→ Use when $V_2 \approx V_1$ (really V_2 is our "threshold voltage")

Avoid toggling $V_{OL} \leftrightarrow V_{OH}$ unnecessarily



V_2 depends on V_O & V_{ref} .

Since positive feedback, assume unstable

→ $V_O = V_{OL}$ or V_{OH}

∴ V_2 takes two possible values:

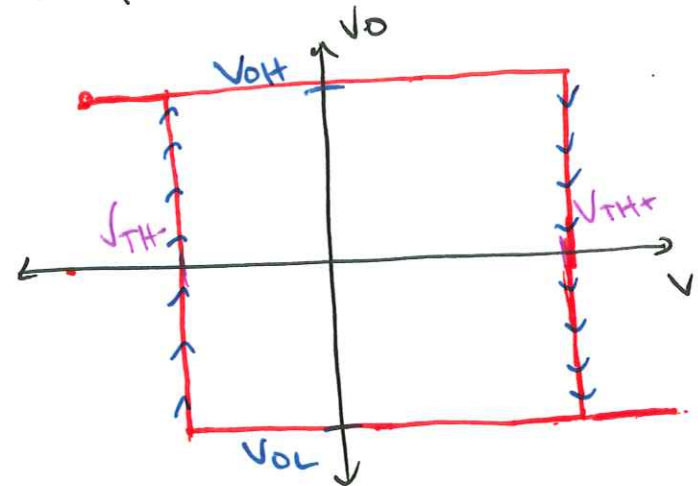
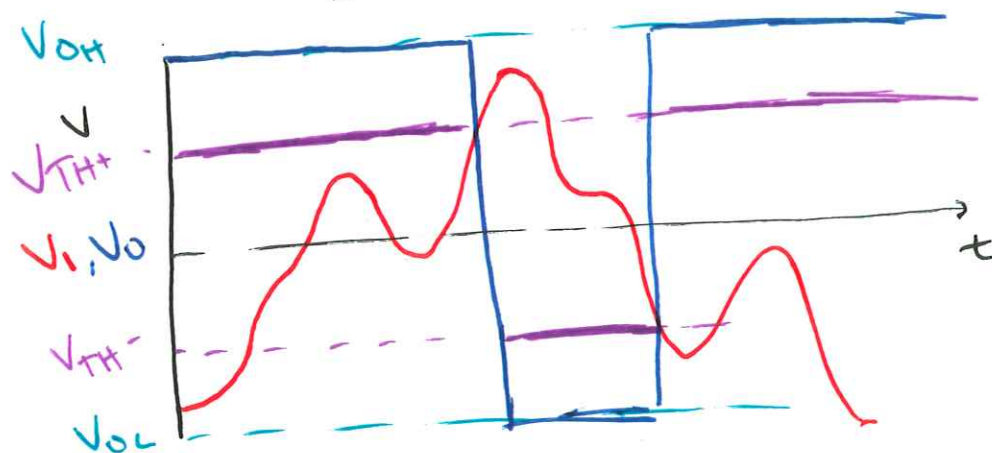
$$V_2 = \begin{cases} (V_{OL} - V_{ref}) \frac{R_1}{R_1 + R_2} + V_{ref} & V_O = V_{OL} = V_{TH-} \\ (V_{OH} - V_{ref}) \frac{R_1}{R_1 + R_2} + V_{ref} & V_O = V_{OH} = V_{TH+} \end{cases}$$

If $V_2 = V_{TH-}$, $V_O = V_{OL}$

V_O will transition to V_{OH} if $V_2 > V_1 \rightarrow V_1 < V_{TH-}$

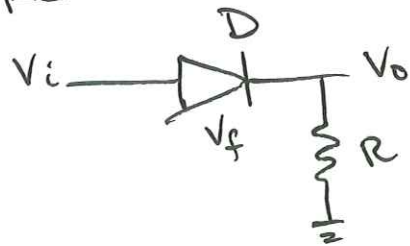
If $V_2 = V_{TH+}$, $V_O = V_{OH}$

" V_{OL} if $V_2 < V_1 \rightarrow V_1 > V_{TH+}$ " "Hysteresis"



2) Add Non-linear Components.

Rectifier:

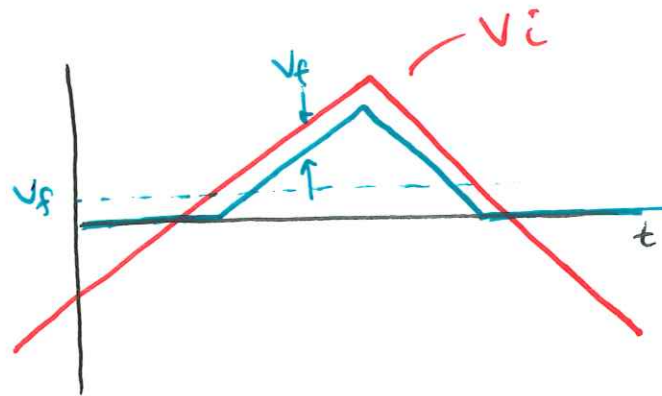


ideal case: $V_f = 0$

$$V_o = \begin{cases} 0 & \text{if } V_i \leq 0 \\ V_i & \text{if } V_i > 0 \end{cases}$$

Non-ideal: $V_f \neq 0$

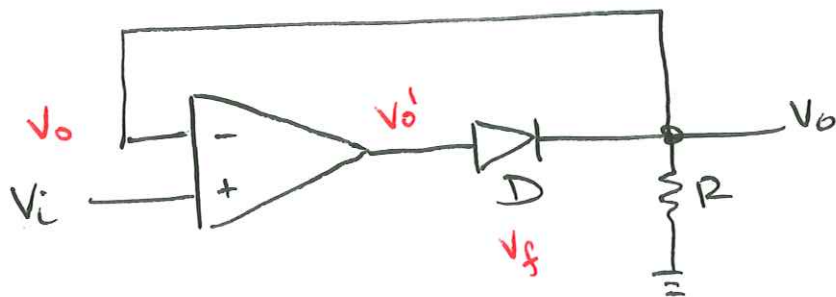
$$V_o = \begin{cases} 0 & \text{if } V_i \leq V_f \\ V_i - V_f & \text{if } V_i \geq V_f \end{cases}$$



fix V_f drop with op amp.



Ideal Rectifier



if $V_i < 0$:

$V_o' < 0$, D is OFF

$V_o \rightarrow 0$ due to R

$\rightarrow V_o' = V_{OL}$

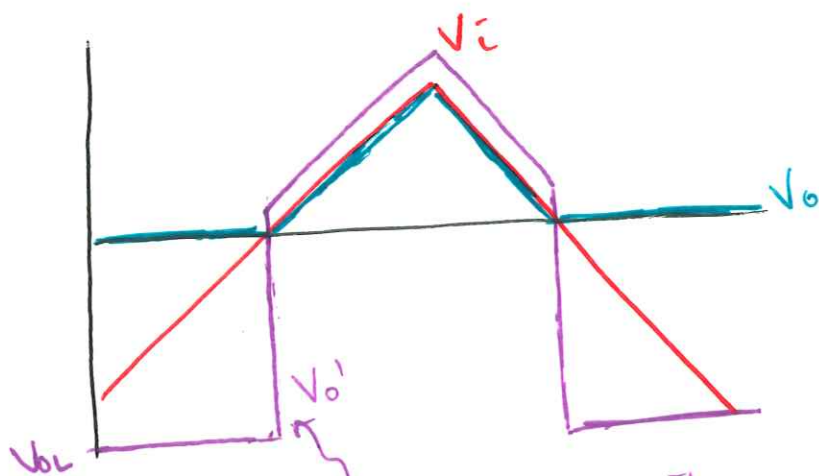
if $V_i > 0$:

$V_o' > 0$, D is ON

$V_o \rightarrow V_i$

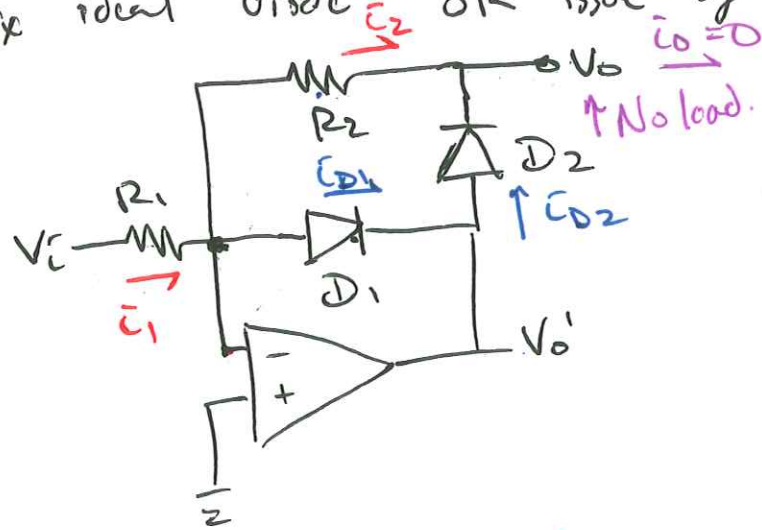
$\rightarrow$ In this case:

$$V_o' = V_o + V_f$$



this transition will be limited by Slew Rate.
Exacerbated by large jump from V_{OL} to V_f

fix ideal diode SR issue by adding additional negative feedback.



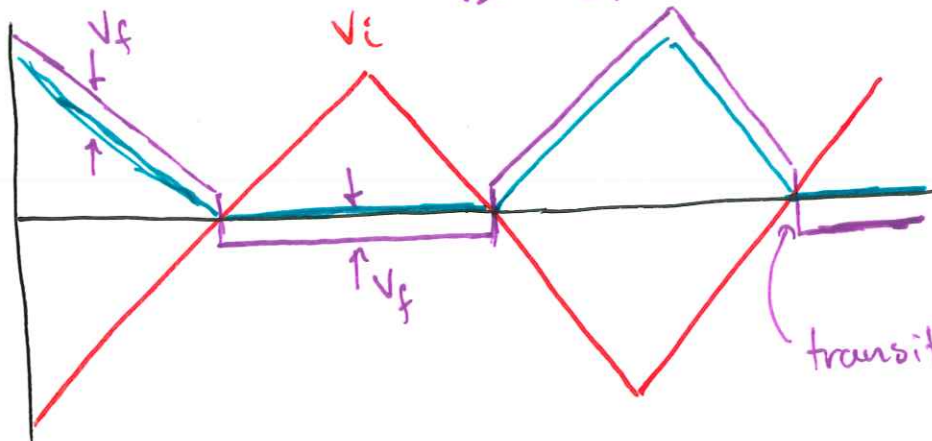
- Two methods to analyze:
- ① Assume virtual short exists, find when invalid.
 - ② Assume $V_n > V_p$ & analyze (saturation)

Method ①: $V_n = 0 = V_p$

if $V_i > 0 \rightarrow i_1 > 0$ implies $i_2 = 0$, $V_o = 0V$
 $i_1 = i_{D1}$, D_1 is ON, $V_o' = -V_f$

if $V_i < 0 \rightarrow i_1 < 0$ implies D_1 is OFF, $i_{D1} = 0$
 $\therefore i_1 = i_2 = i_{D2}$, D_2 is ON, $V_o' = V_o + V_f$

$$i_1 = i_2 \rightarrow -\frac{V_i}{R_1} = \frac{V_o}{R_2} \rightarrow V_o = -\frac{R_2}{R_1} V_i$$



let $R_2 = R_1 \rightarrow V_o = -V_i$ for active region.

Inverted Rectification!

transition if $2V_f$